Google Cloud H2S

Gen AI Academy

APAC Edition

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Participant Details

a. Participant Name: Mithun Raj M R
b. Problem Statement: DecisionIQ: AI-Powered Grocery Inventory Decision Intelligence

Concept Overview: DecisionIQ Platform

DecisionIQ is an AI-powered inventory decision intelligence platform for small grocery store owners. It replaces static sales dashboards and generic chatbots with a dynamic simulation loop centered on a Purchasing Priority Slider:

- Risk Minimization Slider: Toggles priority from Waste reduction (Conservative) to Profit maximization (Aggressive).
- Multi-Scenario Recalculation: Renders and re-ranks three distinct ordering plans dynamically in under 2 seconds.
- Grounded Operations: Outputs 3 immediate, store-level next steps and logs all confirmed decisions locally.
- Fully Deployed: Production-ready on Google Cloud (Cloud Run & Firebase Hosting) connected to live BigQuery datastores.

Technical Approach: Contextual Grounding & Risk Evaluation

• Operational Store (Google BigQuery): Serves as the real-time source of truth for stock levels, unit cost, prices, and shelf lives to eliminate synchronization latency.

• Context Engine: Dynamically fetches localized weather forecasts and event markers (e.g., weekend farmers markets) to forecast impending demand shifts.

• Simulation Core (Vertex AI Gemini 2.5): Formulates structured JSON prompts combining stock stats and environmental context to weigh spoilage write-offs against missed stockouts.

• Actionable Loop: Renders strategy cards, generates plain-English trade-off reasoning (XAI), and saves historical decisions locally.

GCP-Native Integration: Translating database stock, shelf-life, and cost parameters into optimal purchasing decisions.

Opportunities & Unique Selling Propositions (USP):

• Simulation over Static Reporting: While standard dashboards only display historical sales, DecisionIQ actively simulates future scenarios based on risk tolerances and live weather variables.

• Explainable AI (XAI): Eradicates the AI 'black-box' trust barrier by translating raw model parameters into plain-English reasoning (e.g. justifying order hikes based on event forecasts).

• Zero-Overhead Persistence: Saves the last 5 confirmed decisions locally in browser storage, allowing operators to drill down into historical details with no extra database costs.

List of features offered by the solution

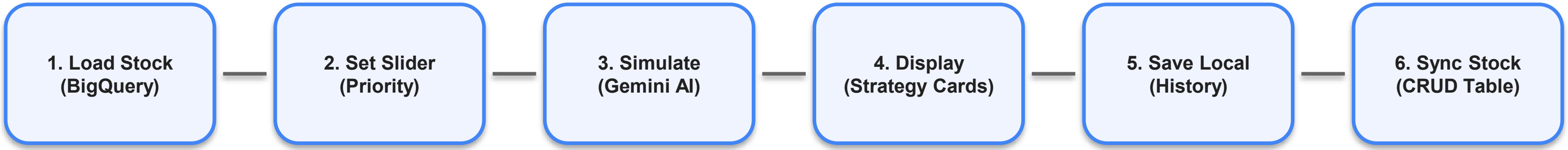
Core Features Specs:

- Priority Slider: Shifts strategy profile (Waste Minimization <-> Profit Maximization) instantly.
- Three Strategy Cards: Compares Conservative, Aggressive, and AI Recommended costs, units, and spoilage risks side-by-side.
- Grounded Next Actions: Lists 3 store-level tasks matching simulated weather/event contexts.
- Local Decision History: Persists confirmed actions chronological timeline inside local browser cache.
- Inline BigQuery CRUD: Directly inserts, updates, and deletes inventory rows with live SQL DML queries.
- Connection Status Console: Displays real-time connectivity status of Vertex AI and BigQuery.

Process Flow: Decision Pipeline Workflow

Decision IQ Process Pipeline Overview:

1. Load Data: Queries current stock levels from BigQuery and tomorrow's weather/event context.
2. Set Priority: Owner slides from Waste to Profit, sending a structured payload to the backend.
3. Run Gemini: Vertex AI Gemini models strategies, calculates optimal units, costs, and risks.
4. Select & Confirm: Renders ranked choices, and saves selection locally in browser timeline.

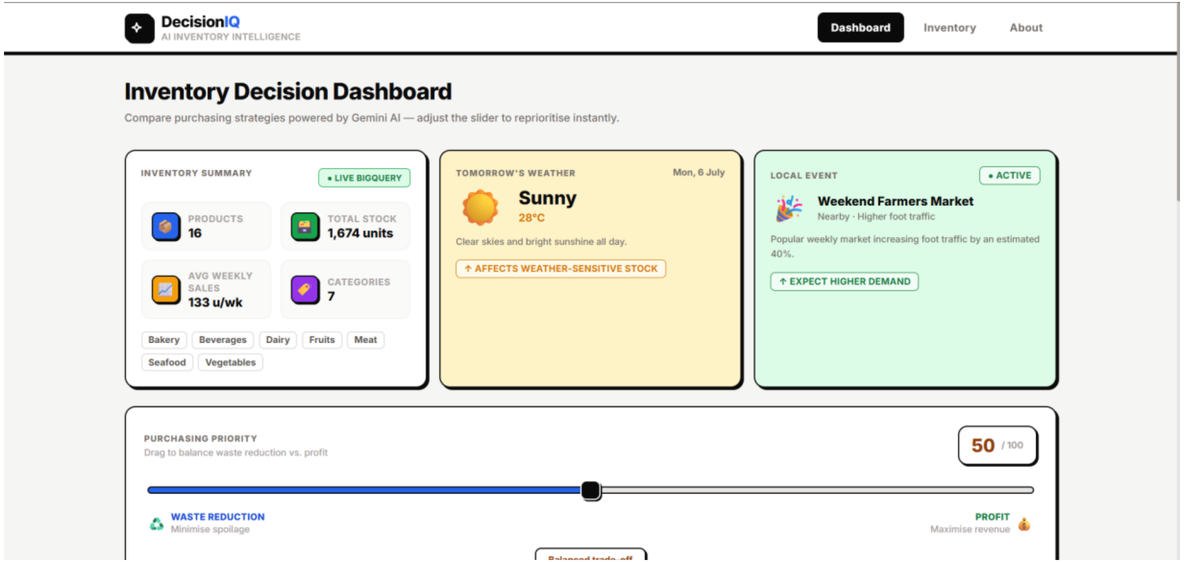


Interactive UI Design: Optimized for Store Floor

Interactive Operator UI:

• High Readability: Designed with heavy ink borders and black offsets to remain clear under bright grocery store lighting.

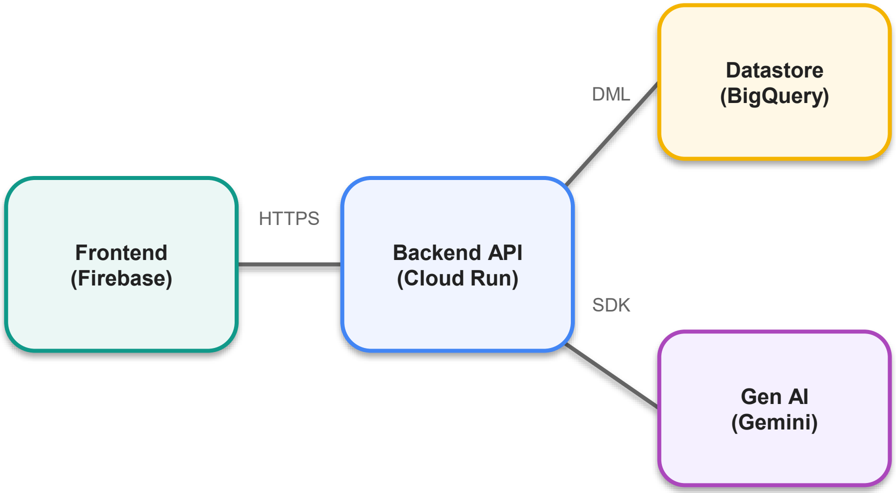
• Dynamic Simulation Dashboard: Renders strategy cards, weather/event contexts, and explainable AI summaries.



Google Cloud Architecture & System Design

Serverless, Cloud-Native Framework:

• Frontend Edge CDN: Deployed on Firebase Hosting for rapid static asset delivery.
• Serverless API Host: FastAPI backend hosted on Google Cloud Run, autoscaling to zero when idle.
• Primary Store: Google BigQuery stores stock, costs, and shelf-life items, executing ACID DML writes.
• AI Coprocessor: Vertex AI Gemini 2.5 maps parameter context to explainable decisions.
• Security: Application Default Credentials (ADC) handle IAM permissions without hardcoded keys in production.



Technologies Used: AI Stack Rationale

Stack Selection & Real-World System Design Rationale:

• Vertex AI Gemini 2.5 Flash: Chosen for low latency and native JSON outputs, keeping slider recalculations under 2 seconds.
• Google BigQuery: Selected as an analytical datastore to query, filter, and aggregate stock items on-the-fly via the Python BigQuery SDK.
• Google Cloud Run: Chosen for lightweight Docker isolation, serverless scale-to-zero cost optimization, and native /health checks.
• Firebase Hosting: Serves the compiled SPA bundle with locked production CORS headers restricting cross-origin access.

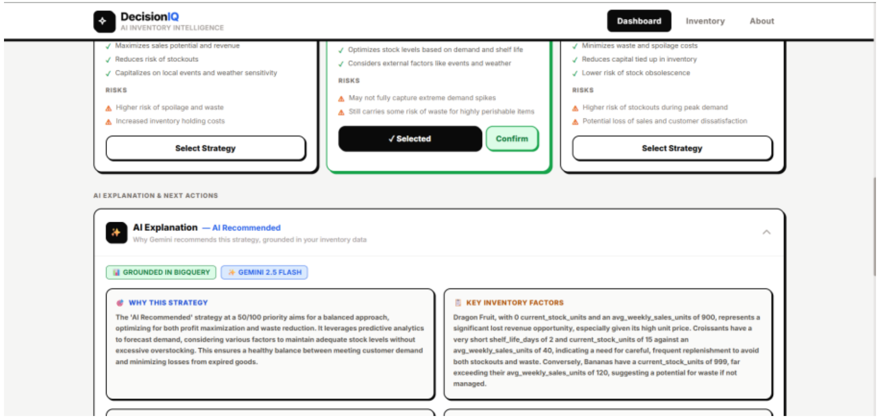
Snapshots of the Live Prototype

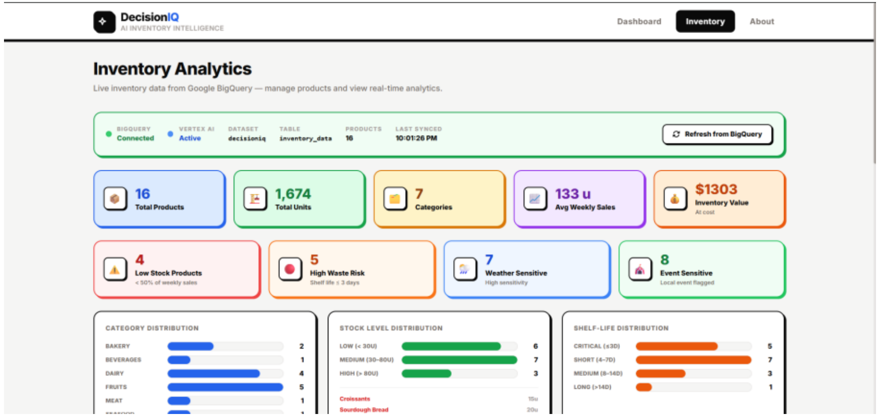
Production Deployed Walkthrough:

• Main Simulation Dashboard: Real-time slider and strategy cards adjusting to priority inputs.
• Inventory Table: Live BigQuery CRUD console with inline edit and delete.
• History Modals: Interactive popups loading weather, events, and AI summaries from localStorage.

GitHub Repository: github.com/mithunrajmr/DecisionIQ

Live Web Application:https://decisioniq-501506-f6f79.web.app





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Thank you